

Lab 4: Decision Tree Classification and Visualization

Aim

To implement a **Decision Tree Classifier** using **Scikit-learn** and visualize the generated decision tree.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate the input features and target values.
4. Split the dataset into training and testing sets.
5. Create a Decision Tree Classifier.
6. Train the model using the training data.
7. Predict the target values for the test data.
8. Calculate and display the accuracy.
9. Visualize the generated decision tree.

Program

```
# Lab 4: Decision Tree Classification and Visualization
```

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
```

```
# Load Iris dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train the model
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Calculate accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Visualize the decision tree
plt.figure(figsize=(12, 8))

plot_tree(
    model,
    feature_names=iris.feature_names,
    class_names=iris.target_names,
    filled=True
)

plt.show()
```

Sample Output

Accuracy: 1.0

A **Decision Tree diagram** will also be displayed showing the feature-based splits used to classify the Iris flowers.

Result

Thus, the **Decision Tree Classifier was successfully implemented using Scikit-learn**, and the generated decision tree was visualized successfully.